IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

APPLIED ADVANCED OPTIMISATION

Thursday, 7 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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APPLIED ADVANCED OPTIMISATION

Information for candidates:

- A function $f : \mathbb{R}^n \to \mathbb{R}$ with **dom** $f = \mathbb{R}^n_{++}$ defined as

$$f(x) = c\, x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n},$$

  with $c > 0$ and $a_i \in \mathbb{R}$, is called a monomial. A sum of monomials of the form

$$f(x) = \sum_{k=1}^{K} c_k x_1^{a_{1k}} x_2^{a_{2k}} \cdots x_n^{a_{nk}},$$

  with $c_k > 0$, is called a posynomial.

- A geometric programme (GP) in posynomial form is described by

$$\begin{array}{ll} \min & f_0(x) \\ s.t. & f_i(x) \leq 1, \quad i = 1, \ldots, m, \\ & h_i(x) = 1, \quad i = 1, \ldots, p, \end{array}$$

  where f_0, ..., f_m are posynomials and h_1, ..., h_p are monomials. A GP in posynomial form can be transformed into convex form by considering the change of variables $y_i = \log x_i$, so $x_i = e^{y_i}$, and then taking the logarithm of all the functions.

- Let $f : \mathbb{R}^n \to \mathbb{R}$. Recall that the conjugate of the function f is defined as

$$f^*(y) = \sup_{x \in \mathbf{dom}\, f} \left(y^\top x - f(x) \right),$$

- Given a function $f : \mathbb{R}^n \to \mathbb{R}$, the Newton step Δx_{nt} is defined as

$$\Delta x_{nt} = -(\nabla^2 f(x))^{-1} \nabla f(x),$$

1. a) A cloud computing centre consists of N servers that must process a total incoming workload of rate L (requests per second). You need to determine the workload x_i (requests per second) assigned to each server i, for $i = 1, \ldots, N$. The workload must be non-negative and the total workload assigned must equal the incoming rate L. Each server i has a maximum service capacity C_i and to ensure stability, the load x_i assigned to server i must be strictly less than its capacity C_i. The system seeks to balance two objectives:

- Minimise latency: the average latency on server i is modelled by the queuing function $d_i(x_i) = \frac{1}{C_i - x_i}$. The total system latency cost is the sum of these latencies.
- Minimise power: the power consumption of server i is modelled as a quadratic function of the load: $p_i(x_i) = \alpha_i x_i^2 + \beta_i x_i$, where $\alpha_i, \beta_i > 0$.

i) Formulate this resource allocation problem as a convex optimisation problem where you minimise the weighted sum of total latency and total power, using a weight parameter $\lambda > 0$ for the power component. [5 marks]

ii) Prove that the function $f(x) = \frac{1}{C-x}$ is convex for $x < C$. [2 marks]

iii) Explain the significance of the parameter λ. Specifically, what does the solution $x^*(\lambda)$ represent in terms of the trade-off between latency and power? [2 marks]

iv) Describe the theoretical problem arising with requiring that the load x_i assigned to server i must be *strictly* less than its capacity C_i rather than using a non-strict inequality. [1 mark]

b) Consider the design of a closed cylindrical beverage can with radius r and height h. The can is manufactured from a thin sheet of metal. The can must hold at least a volume V_{min}. The height h must be at least twice the radius r, but not more than four times the radius. The cost of the can is proportional to its total surface area. The top of the can requires a thicker metal, costing 3 times as much per unit area as the bottom and sides. The height plus the diameter must not exceed a production machine limit L_{max}. You seek to minimise the total material cost.

Hint: the circumference and area of a circle of radius r are $2\pi r$ and πr^2, respectively.

i) Formulate this design problem as a Geometric Programme (GP) in posynomial form. [5 marks]

ii) Transform the Geometric Programme derived in part (i) into a convex optimisation problem. Use the variables $y_r = \log r$ and $y_h = \log h$.

[5 marks]

2. Consider the maximum entropy problem:

$$\begin{aligned} \min_{x} \quad & \sum_{i=1}^{n} x_i \log x_i \\ \text{s.t.} \quad & \sum_{i=1}^{n} x_i = 1 \end{aligned}$$

where the domain of the objective function is $\mathbb{R}^n_{++}$ (i.e., $x_i > 0$ for all i).

a) Let $f(x) = x \log x$ be a scalar function defined on $x > 0$. Show that its conjugate function is given by:

$$f^*(y) = e^{y-1}$$

You must provide all steps of the derivation. [4 marks]

b) Using the result from part (a), derive the Lagrange dual function $g(\nu)$ for the entropy problem stated above, where $\nu \in \mathbb{R}$ is the dual variable associated with the equality constraint.

Hint: Recall that $\inf_x(h(x) + y^T x) = -\sup_x(-y^T x - h(x)) = -h^*(-y)$*, where* $h^*(y)$ *is the conjugate function of* $h(x)$ [6 marks]

c) State the full set of Karush-Kuhn-Tucker (KKT) conditions for this problem.

[4 marks]

d) Consider the specific simple case where $n = 2$.

i) Using the KKT conditions, find the values of x_1^* and x_2^* that minimise the objective. Calculate the optimal primal value p^*. [3 marks]

ii) Analytically find the optimal dual variable ν^* by maximising the dual function $g(\nu)$ obtained in part (b). Calculate the optimal dual value d^*. Does strong duality hold? [3 marks]

3. a) Answer the following convexity questions.

i) Determine whether $f(x) = x^T x$ defined on $\mathbb{R}^n$ is convex, concave, or neither. Justify your answer. [2 marks]

ii) Determine whether $f(x_1, x_2) = \frac{x_1^2}{x_2}$ defined on $\mathbb{R} \times \mathbb{R}_{++}$ is convex, concave, or neither. Prove your answer using the Hessian. [2 marks]

iii) Determine whether $f(x) = \log(e^{x_1} + e^{x_2})$ defined on $\mathbb{R}^2$ is convex, concave, or neither. Prove your answer using the Hessian. [2 marks]

iv) Let S_1 and S_2 be convex sets in $\mathbb{R}^n$. Prove that their intersection $S_1 \cap S_2$ is a convex set using the definition of a convex set. [2 marks]

b) Find the dual function of the LP

$$\begin{array}{ll} \min & c^\top x \\ s.t. & x \succeq 0 \\ & Ax = b. \end{array}$$

Write the dual problem, make the implicit equality constraints explicit, and eliminate the Lagrange multipliers for the inequality constraints.

[4 marks]

c) Consider the strictly convex quadratic minimisation problem:

$$\min_x \quad f(x) = \frac{1}{2} x^T P x + q^T x + r$$

where $P \in \mathbb{R}^{n \times n}$ is symmetric and positive definite ($P \succ 0$), $q \in \mathbb{R}^n$, and $r \in \mathbb{R}$.

i) Derive the expression for the Newton step Δx_{nt} at an arbitrary point x. [3 marks]

ii) Let $x^{(0)}$ be any starting point. Show that the next iterate $x^{(1)} = x^{(0)} + \Delta x_{nt}$ is the exact global minimiser of $f(x)$. [3 marks]

iii) Explain why the Gradient Descent typically fails to converge in one step for such a problem, referring to P and how this may influence the geometry of the level sets. [2 marks]